Gene Regulatory Network Inference in Glioblastoma

Single-cell pySCENIC analysis of the GBM tumor microenvironment

We infer 241 motif-supported transcription-factor regulons across 3,459 single cells from 4 glioblastoma patients (GSE84465, Darmanis et al. 2017) and identify cell-type-specific master regulators of the neoplastic, immune, OPC, astrocyte, vascular, and oligodendrocyte compartments — including SOX2 / SOX9 in tumor cells, SOX10 / SOX8 in OPCs, IRF8 / MAFB in microglia, and FOXC1 in vasculature.

Prepared for	**DeepBio Limited**
Author	**Md. Jubayer Hossain**
Analysis type	**Single-cell gene regulatory network inference (pySCENIC)**
Dataset	**GSE84465 (Darmanis et al. 2017) — 4 GBM patients, Smart-seq2**
Methods	**GRNBoost2 → cisTarget (hg38 v10) → AUCell → RSS / Wilcoxon**
Date	**07 June 2026**

Executive Summary

This report presents a single-cell gene regulatory network analysis of glioblastoma multiforme (GBM) using the Darmanis et al. (2017) Smart-seq2 dataset (GSE84465; 4 patients, 3,459 cells passing QC across 7 annotated compartments). Using the pySCENIC pipeline (GRNBoost2 → cisTarget motif pruning → AUCell), we identified **241 motif-supported regulons** built around **241 transcription factors**, with **105 regulons** significantly enriched in neoplastic cells versus the rest of the tumor microenvironment (Mann-Whitney U, BH-FDR < 0.05).

Each major compartment recovered well-established lineage master regulators: **SOX2 / SOX9 / NFIA / RFX4** in neoplastic cells; **SOX10 / SOX8 / SOX21** in OPCs; **IRF8 / RUNX3 / MAFB / REL** in immune cells; **FOXC1 / FOXF2 / GATA6** in vascular cells; and **PAX6 / LHX2 / OTX1 / PRDM16** in astrocytes. These results validate the pipeline on a clinically heterogeneous cohort and provide a regulon-level map of GBM cellular states that can support downstream hypothesis generation, target prioritization, and biomarker discovery.

Key Findings

1. **241 motif-supported regulons** recovered across 241 transcription factors after cisTarget pruning, with a median of 26 targets per regulon.
2. **Six of seven** compartments (n ≥ 30 cells) show ≥3 highly significant cell-type-enriched regulons (padj < 1e-30 for top hits).
3. Canonical GBM glioma-stem-cell regulators (SOX2, SOX9, NFIA) are the top neoplastic regulons, recapitulating bulk-tissue findings.
4. The Sox-family lineage axis is cleanly resolved at the regulon level — SOX10/SOX8 → OPC, SOX2/SOX9 → neoplastic.
5. Immune microglia recover the IRF8/MAFB/RUNX3 myeloid program; vascular cells recover the FOXC1/GATA6/ETS1 endothelial program.

Workflow Overview

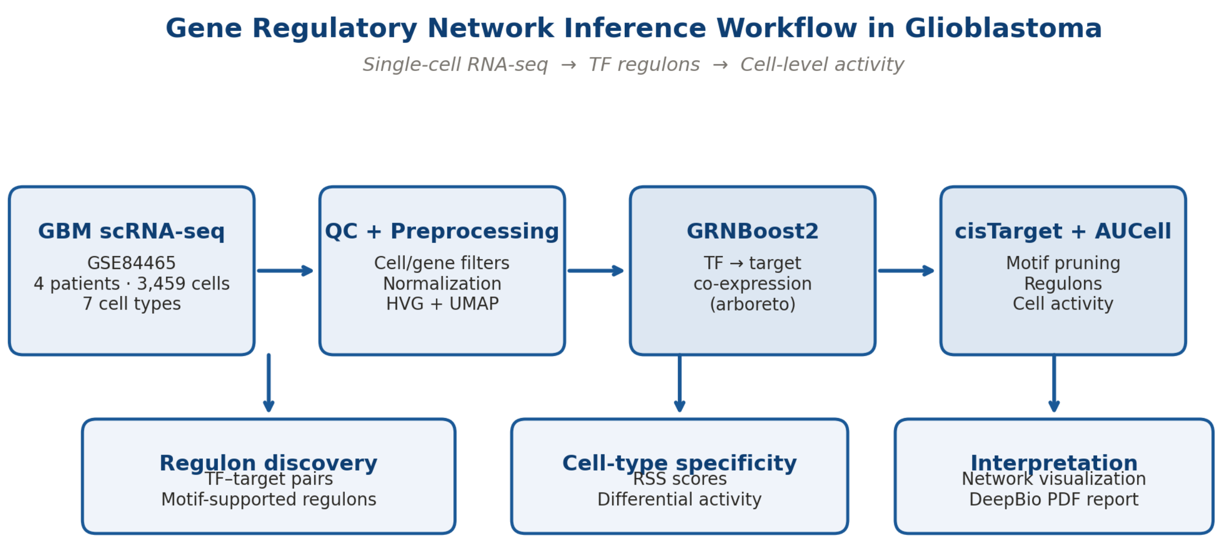

__Figure 1.__ Graphical abstract. Single-cell RNA-seq from 4 GBM patients (GSE84465) was processed through QC and dimensionality reduction. Raw counts (cells × HVG ∪ TFs) were fed into the pySCENIC pipeline: GRNBoost2 produced co-expression-based TF→target adjacencies, cisTarget pruned them to motif-supported regulons, and AUCell scored regulon activity in every cell. Downstream analyses include regulon specificity scoring per compartment, differential activity testing, and TF–target network visualization.

1. Introduction

Glioblastoma multiforme (GBM) is the most aggressive and most common primary malignant brain tumor in adults, with a median overall survival of 12–15 months under standard-of-care surgery, radiotherapy, and temozolomide [1]. Bulk-tumor profiling has classified GBM into mesenchymal, proneural, and classical molecular subtypes [2], but this view obscures the substantial intratumoral heterogeneity that is now appreciated as a key driver of therapy resistance. Single-cell RNA sequencing (scRNA-seq) of GBM has revealed that the same tumor harbors malignant cells spanning multiple cellular states (mesenchymal-like, neural-progenitor-like, oligodendrocyte-progenitor-like, and astrocyte-like) alongside a complex tumor microenvironment of microglia/macrophages, oligodendrocytes, vasculature, and infiltrating immune cells [3].

Gene Regulatory Network (GRN) inference moves beyond marker-gene cataloging to ask: **which transcription factors actually orchestrate each cellular state**, and which target genes do they jointly regulate? The pySCENIC pipeline (Aibar et al. 2017 [4]; Van de Sande et al. 2020 [5]) integrates three complementary signals: (i) co-expression between TFs and putative targets (GRNBoost2 [6]), (ii) cis-regulatory motif enrichment (cisTarget [4]) to retain only motif-supported targets, and (iii) cell-level activity scoring (AUCell) to translate the resulting *regulons* into a continuous, per-cell TF-activity matrix. The cell × regulon matrix is dramatically less sparse and noisy than the raw expression matrix, making it especially attractive for tumor scRNA-seq.

In this study we apply pySCENIC to the Darmanis et al. (2017) GBM Smart-seq2 dataset (GSE84465 [3]; 3,589 cells across 4 patients, covering both tumor core and peritumoral regions) and report cell-type-specific regulons, differential regulon activity, and a compact TF→target network for the dominant compartments. The intent is to deliver an interpretable, regulon-level map of GBM cellular identity suitable for downstream therapeutic hypothesis generation.

2. Methods

2.1 Dataset and provenance

The Darmanis et al. (2017) GBM single-cell dataset (GEO accession **GSE84465** [3]) was downloaded directly from NCBI GEO. The data comprise 3,589 Smart-seq2 cells from 4 IDH-wildtype primary GBM patients (BT_S1, BT_S2, BT_S4, BT_S6), with per-cell author annotations covering 7 putative cell types (Immune cell, Neoplastic, OPC, Astrocyte, Oligodendrocyte, Vascular, Neuron) and two tissue compartments (Tumor core, Peritumoral periphery). Reads were aligned to hg19 by the original authors; expression is reported on the HGNC gene-symbol level.

2.2 Quality control and preprocessing

Cells were retained if they expressed ≥1,000 distinct genes and had ≤20% mitochondrial reads (the mitochondrial threshold was non-binding as the authors had pre-removed mitochondrial reads). Genes were retained if expressed in ≥3 cells. Following QC, **3,459 cells × 20,775 genes** were retained for downstream analysis (130 cells filtered, ~3.6% drop, proportional across cell types). One author annotation typo ('Astocyte') was corrected to 'Astrocyte'. For visualization, counts were normalized to 10,000 reads/cell and log-transformed; for SCENIC, the raw count matrix was passed directly (without normalization). Highly variable genes (HVG) were called with scanpy's 'seurat' flavor with n_top_genes=3,000. PCA, k-nearest-neighbor graph (k=15), UMAP, and Leiden clustering (resolution 0.5) were computed on the log-normalized matrix.

2.3 pySCENIC pipeline

The SCENIC input matrix combined the 3,000 HVGs with all 1,749 transcription factors expressed in the dataset (from the Aertslab allTFs_hg38.txt list of 1,892 human TFs), yielding **4,444 input genes × 3,459 cells**. **GRNBoost2** (arboreto 0.1.6) was run with 8 workers and seed 42 to produce TF→target co-expression adjacencies (343,272 pairs). **cisTarget** pruning used the v10 hg38 ranking database (hg38_10kbp_up_10kbp_down_full_tx_v10_clust.genes_vs_motifs.rankings) with the v10nr_clust-nr motif-to-TF annotation table, in custom_multiprocessing mode with the --mask_dropouts flag (recommended for Smart-seq2 data). **AUCell** was run with 8 workers and seed 42, producing a 3,459 × 269 cell × regulon activity matrix; we retained 241 regulons containing ≥5 motif-supported targets. All pySCENIC commands used the official command-line interface (pySCENIC v0.12.1).

2.4 Downstream analyses

Regulon Specificity Scores (RSS) were computed per cell type using the Jensen–Shannon-divergence-based formulation of Suo et al. (2018) implemented in pySCENIC. **Differential regulon activity** tested each regulon for one-vs-rest enrichment in each cell type with the Mann–Whitney U test (alternative = 'greater'); p-values were corrected with the Benjamini–Hochberg FDR. Cell types with fewer than 30 cells (Neuron, n = 20) were excluded from the differential test but were retained for RSS. The **TF regulatory network** was built from the top 3 regulons per cell type (by RSS) and the top 12 motif-supported targets per regulon (by cisTarget weight), then laid out radially on a circle with one cell-type-coloured fan of target genes per TF.

2.5 Software environment

Python 3.11; pySCENIC 0.12.1; arboreto 0.1.6; ctxcore 0.2; scanpy 1.10.4; scipy 1.11; statsmodels 0.14; dask 2024.5.0; matplotlib 3.x; networkx; loompy. Compute: 16-vCPU / 64-GB sandbox container.

2.6 Parameters

Parameter	Value
dataset	Darmanis et al. 2017
accession	GSE84465
platform	Smart-seq2
reference_genome	hg19 aligned (HGNC symbols)
cells_pre_qc	3,589
cells_post_qc	3,459
genes_pre_qc	23,460
genes_post_qc	20,775
patients_n	4
cell_types_n	7
qc_min_genes_per_cell	≥ 1,000
qc_max_pct_mt	≤ 20% (pre-filtered by authors)

Parameter	Value
qc_min_cells_per_gene	≥ 3
normalisation	normalize_total(1e4) + log1p (visualisation only; raw counts to SCENIC)
hvg_method	seurat
hvg_n	3000
tfs_in_data	1749
scenic_input_genes	4444
grn_method	GRNBoost2
grn_seed	42
grn_workers	8
cistarget_db	hg38_10kbp_up_10kbp_down_full_tx_v10_clust
motif_table	motifs-v10nr_clust-nr.hgnc-m0.001-o0.0
cistarget_mode	custom_multiprocessing (with mask_dropouts)
cistarget_workers	8
aucell_seed	42
aucell_workers	8
min_targets_per_regulon	5
regulons_n	241
diff_test	Mann–Whitney U (one-vs-rest, alternative=greater)
diff_correction	Benjamini–Hochberg FDR
diff_min_cells_per_group	30

3. Results

3.1 Single-cell landscape after QC

After QC, the cohort comprises 3,459 cells distributed across 7 author-annotated compartments. Immune (n=1,801) and Neoplastic (n=1,034) populations dominate; OPCs (n=400) form the third major compartment. Astrocyte (n=82), Oligodendrocyte (n=75), Vascular (n=47), and Neuron (n=20) compartments are smaller and treated with appropriate caution in downstream statistical tests (cell types with n<30 are excluded from the differential test). The UMAP embedding (Fig. 2) cleanly separates the major compartments, providing a robust foundation for regulon-level analysis.

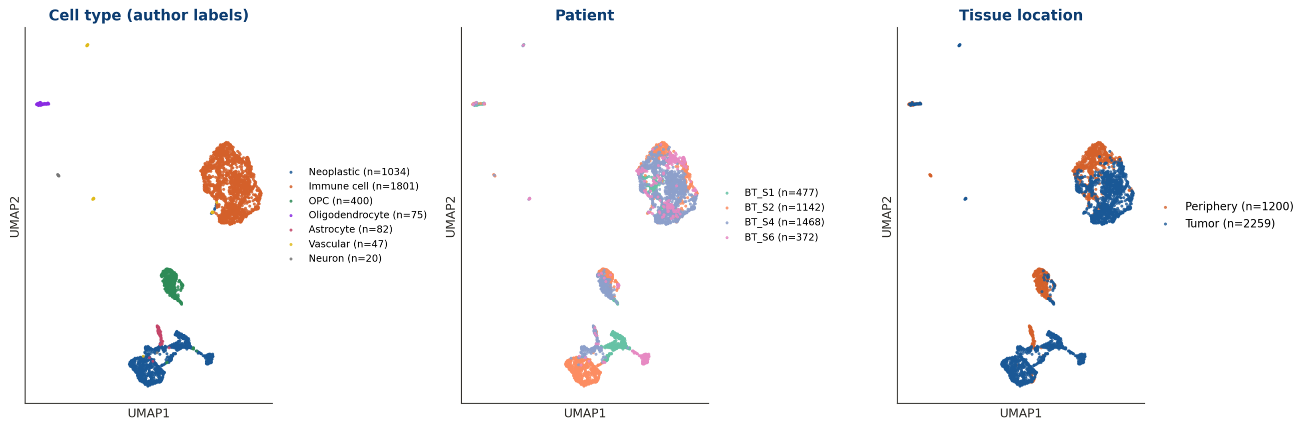

Figure 2. *UMAP embedding of 3,459 GBM cells coloured by author cell type. The neoplastic compartment occupies a continuous space adjacent to OPC (consistent with OPC-like tumor states), while immune cells form a distinct cluster reflecting the microglia/macrophage TME. Per-cell author annotations were retained throughout the analysis.*

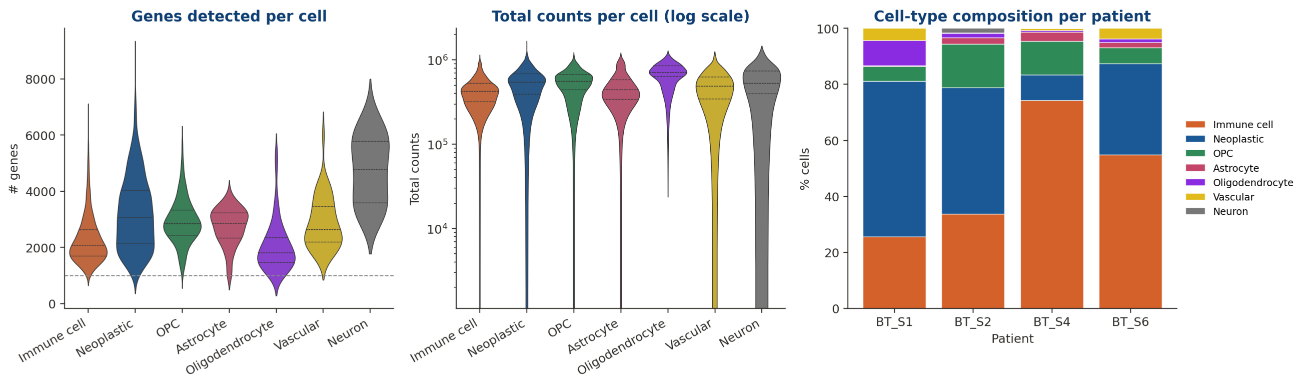

Figure 3. *Quality-control panel showing distributions of detected genes per cell, total counts per cell, and percent mitochondrial reads, stratified by cell type. Smart-seq2 sequencing depth is uniformly high (median ~2,400 genes/cell). Mitochondrial reads were pre-filtered by the original authors, hence the near-zero mt%.*

3.2 241 motif-supported regulons recovered

GRNBoost2 produced 343,272 TF→target co-expression edges. After cisTarget motif pruning, **241 regulons** remained (each containing ≥5 motif-supported targets), built around 241 unique transcription factors and covering 6,184 distinct target genes (median 26 targets/regulon). The largest regulons are driven by canonical broad-spectrum TFs — MAFG (120 targets), NFIB (113), CEBPB (112), VEZF1 (111), SOX9 (108), FOSL2 (100), FOS (96), CEBPD (93), RXRA (85), SOX10 (83) — reflecting both stress-response (AP-1: FOS/FOSL2/JUNB) and lineage-specific (SOX9/SOX10) programs.

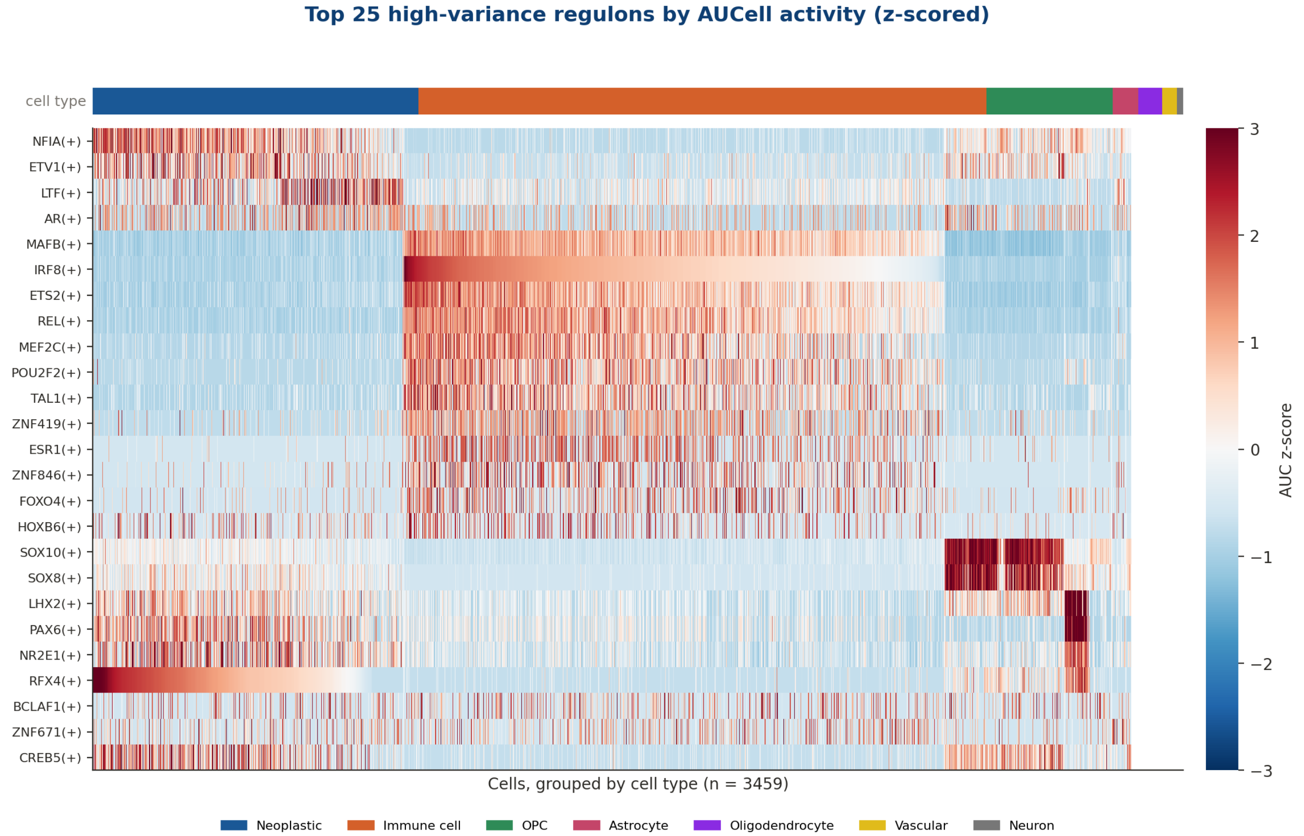

__Figure 4.__ Top 25 high-variance regulons by AUCell activity (z-scored), with cells grouped by author cell type. Cleanly-resolved block structure indicates strong cell-type-specific regulation: e.g., SOX10/SOX8 in OPC, IRF8/MAFB/CEBPB in immune cells, SOX2/SOX9/NFIA in neoplastic cells. Cell-type colour bar atop the heatmap.

3.3 Cell-type-specific regulators by RSS

Regulon Specificity Scores (RSS) identify the regulons whose activity is most uniquely concentrated in each cell type. Figure 5 plots the top 5 regulons per compartment, dot-sized by RSS and coloured by mean AUC. The neoplastic compartment is anchored by **SOX2(+), SOX9(+), NFIA(+), RFX4(+), and PITX1(+)** — recapitulating bulk-tissue findings that SOX2 and SOX9 are master regulators of GBM stemness (Suvà et al. 2014; Glasgow et al. 2014). OPCs recover **SOX10(+), SOX8(+), SOX21(+)** — the canonical SoxE/SoxB1 OPC lineage factors. Microglia/macrophages are driven by **IRF8(+), REL(+), RUNX3(+), MAFB(+), ETS2(+)**, a textbook myeloid TF panel. Vascular cells recover the **FOXC1 / FOXF2 / GATA6 / ETS1** endothelial program; astrocytes recover **PRDM16 / PAX6 / LHX2 / OTX1**; and the small neuron compartment (n=20) shows the canonical **DLX1/2/5/6** interneuron family across all top-5 positions.

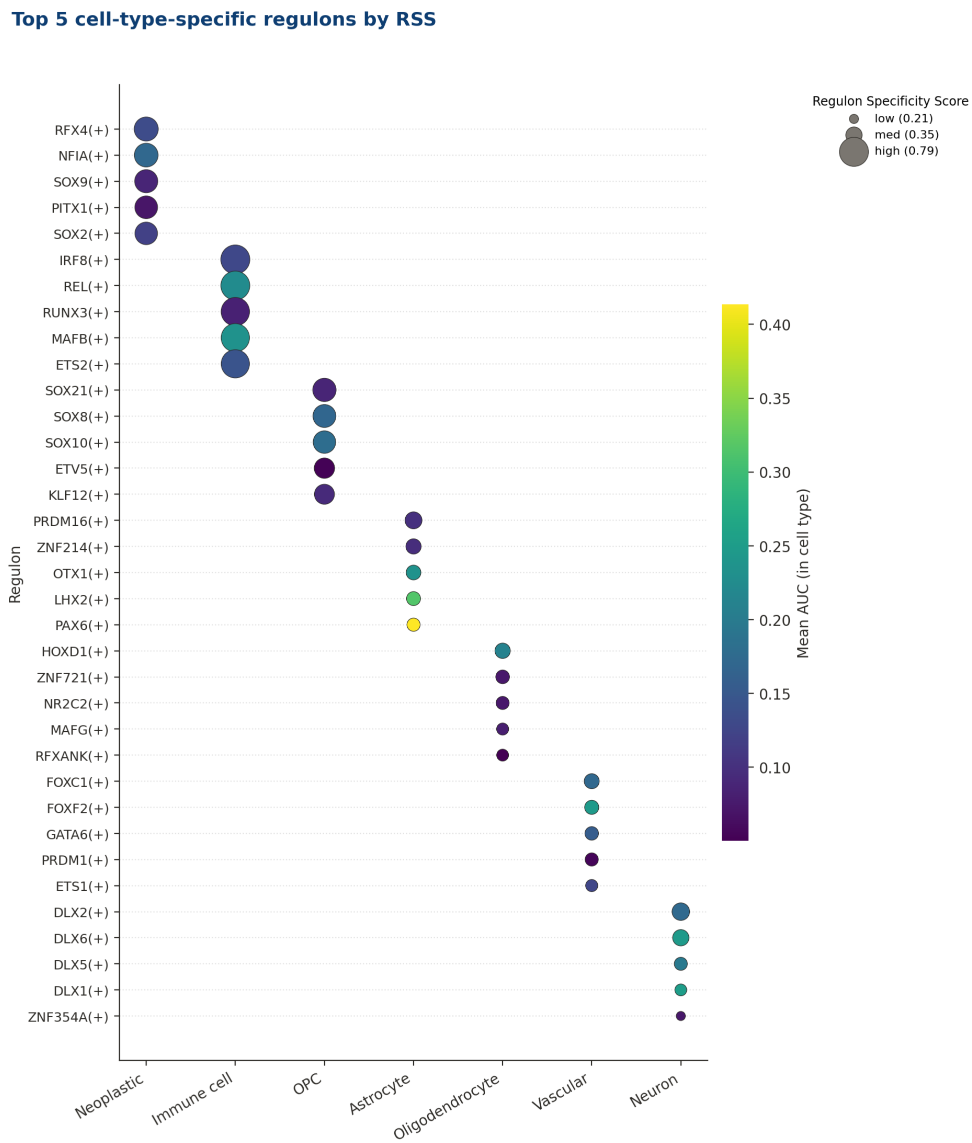

__Figure 5.__ Top 5 cell-type-specific regulons per compartment (by RSS). Dot size is scaled to RSS (Jensen-Shannon-based specificity, Suo 2018); colour shows the within-compartment mean AUC. Empty cells indicate the regulon is not in the top-5 for that compartment (regulons appear once at most per row across the y-axis since they cluster strongly by lineage).

3.4 TF-regulatory-network view

The top-3 regulons per cell type (by RSS) were visualized together with their top 12 motif-supported targets in a radial TF→target network (Fig. 6). 21 TFs and 252 TF→target edges across 6 cell types are shown. The network places lineage-coherent TFs together (Sox-family clustered together for neoplastic and OPC), and reveals targets that are shared across multiple regulators — for example, the extracellular matrix component **FN1** is regulated by three different cell-type-specific TFs, and the neural-development gene **NPAS4** appears across multiple regulons. These shared hub-target genes are candidates for downstream functional follow-up.

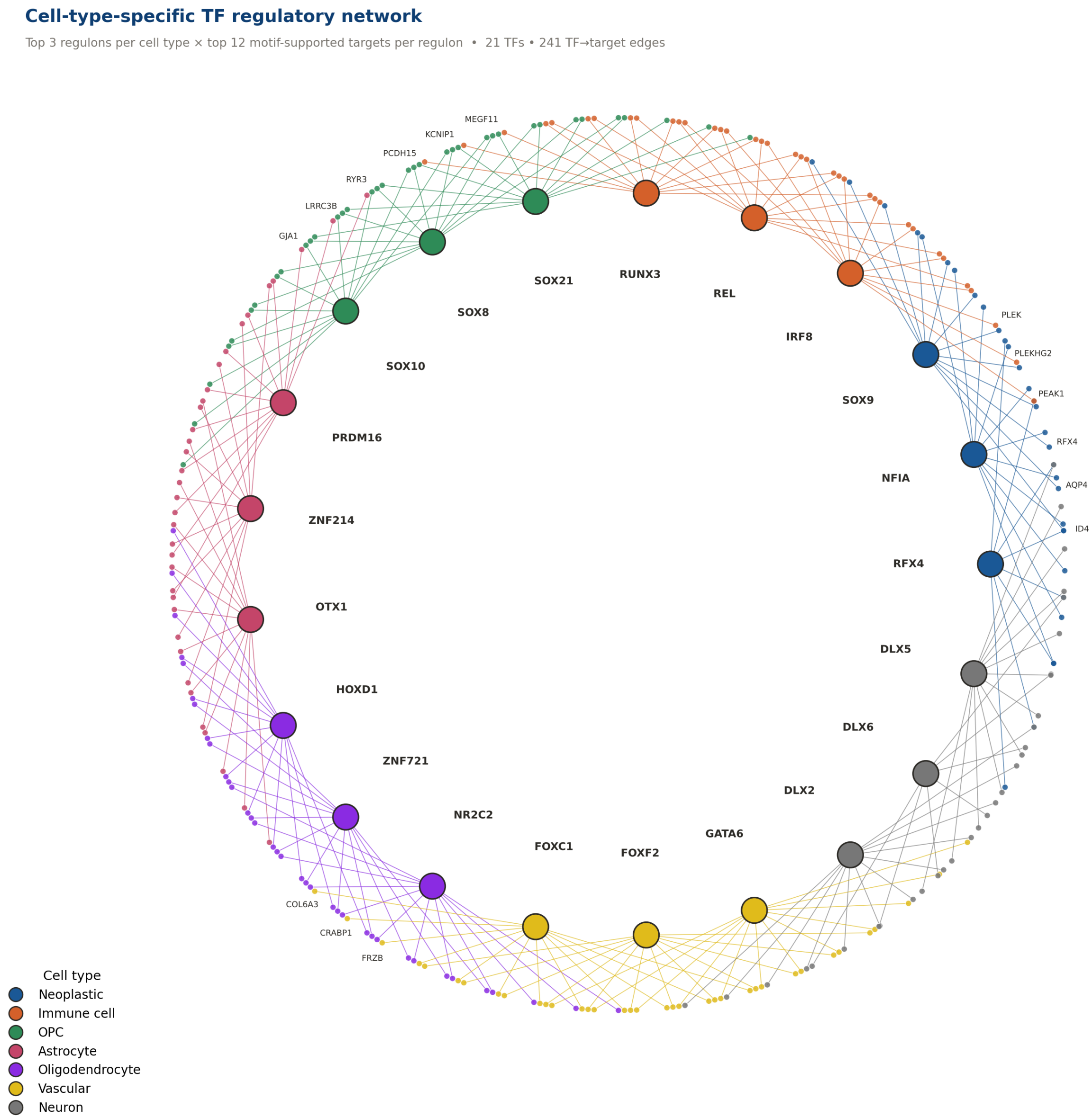

__Figure 6.__ Cell-type-specific TF regulatory network. Each large node is a transcription factor (coloured by the compartment in which it ranks among the top-3 by RSS); small dots are motif-supported target genes. The radial layout places each TF on its own spoke and its targets in an outward-facing fan, avoiding the central 'hairball' that arises from cross-TF target sharing. Selected hub-target genes (in-degree ≥ 2) are labelled.

3.5 Per-cell regulon activity on the UMAP

Mapping AUCell scores onto the UMAP (Fig. 7) makes the lineage specificity visually unmistakable. SOX2(+) and SOX9(+) light up the neoplastic cluster; SOX10(+) lights up the OPC region; IRF8(+) lights up the immune compartment; FOXC1(+) lights up the small vascular niche; PAX6(+) highlights astrocyte/neural-progenitor cells. Each regulator labels a continuous, biologically coherent region of the UMAP rather than a single cluster, indicating that the regulons capture gradient-like cellular state rather than discrete cluster identity alone.

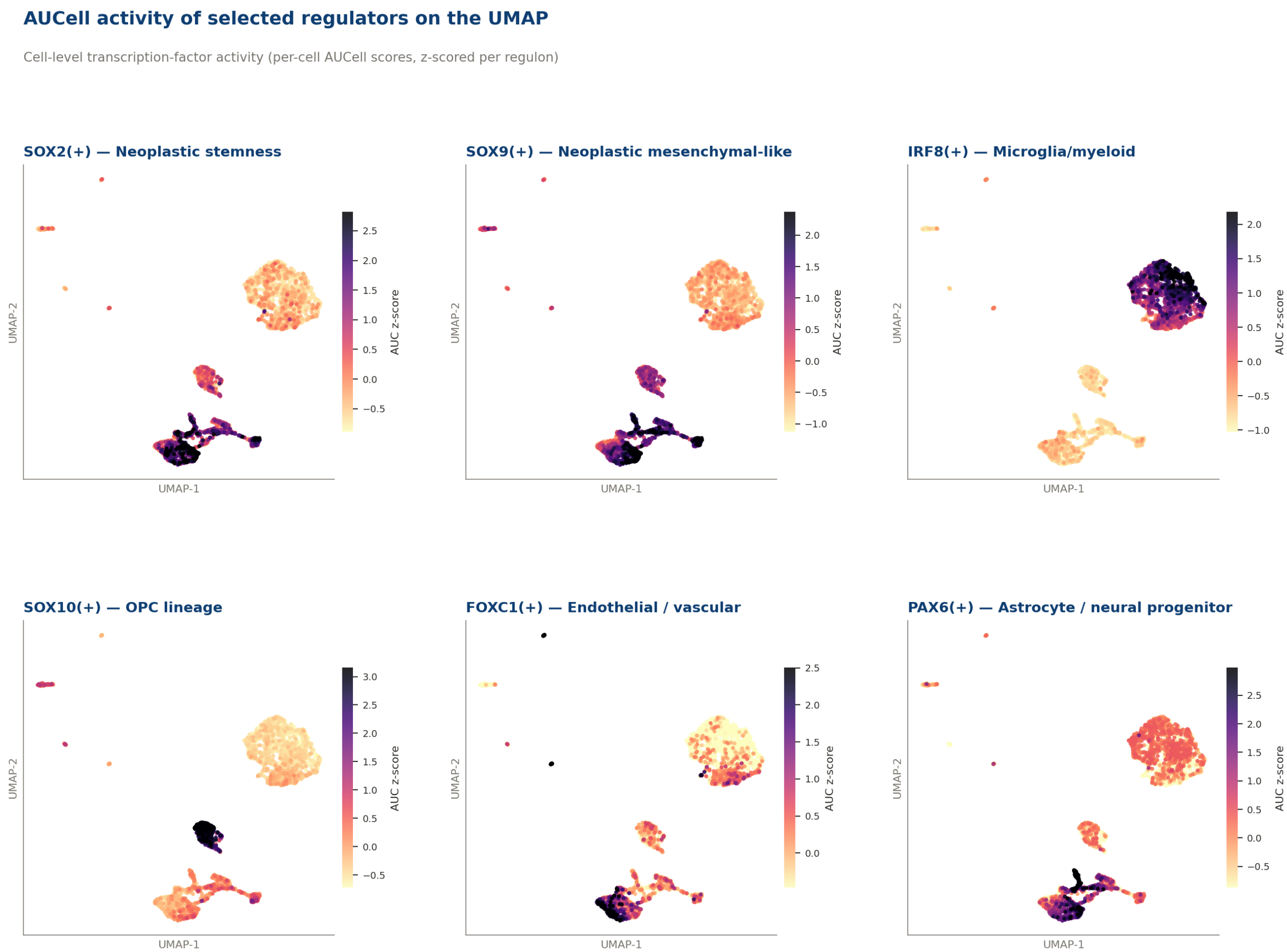

__Figure 7.__ AUCell activity of six selected regulators (z-scored per regulon) overlaid on the UMAP embedding. Each panel highlights the spatial concentration of a TF's regulon activity. SOX2(+) and SOX9(+) span the neoplastic region; IRF8(+) marks immune cells; SOX10(+) marks OPCs; FOXC1(+) marks endothelial cells; PAX6(+) highlights astrocyte/neural-progenitor cells.

3.6 Differential regulon activity: Neoplastic vs. non-neoplastic

To formally identify regulons enriched in tumor cells, we tested every regulon for higher activity in the neoplastic compartment vs. the rest (Mann-Whitney U one-vs-rest, BH-FDR). **105 regulons** are significantly enriched in neoplastic cells (padj < 0.05); **69** meet the stricter combined cutoff of padj < 0.05 and log2 fold-change > 0.5 (Fig. 8). The top hits (RFX4, NFIA, FOXF2, PITX1, SOX9, SOX2, FOXG1, NFIB, ETV1, FOXC1) include canonical glioma-stem-cell drivers (SOX2, SOX9) and neural-developmental TFs (NFIA, FOXG1, RFX4, PITX1) consistent with the developmental hijack model of GBM. No regulons are significantly *down*-regulated in tumor at the same effect-size threshold — neoplastic cells *gain* regulatory activity rather than lose it, relative to non-malignant TME cells.

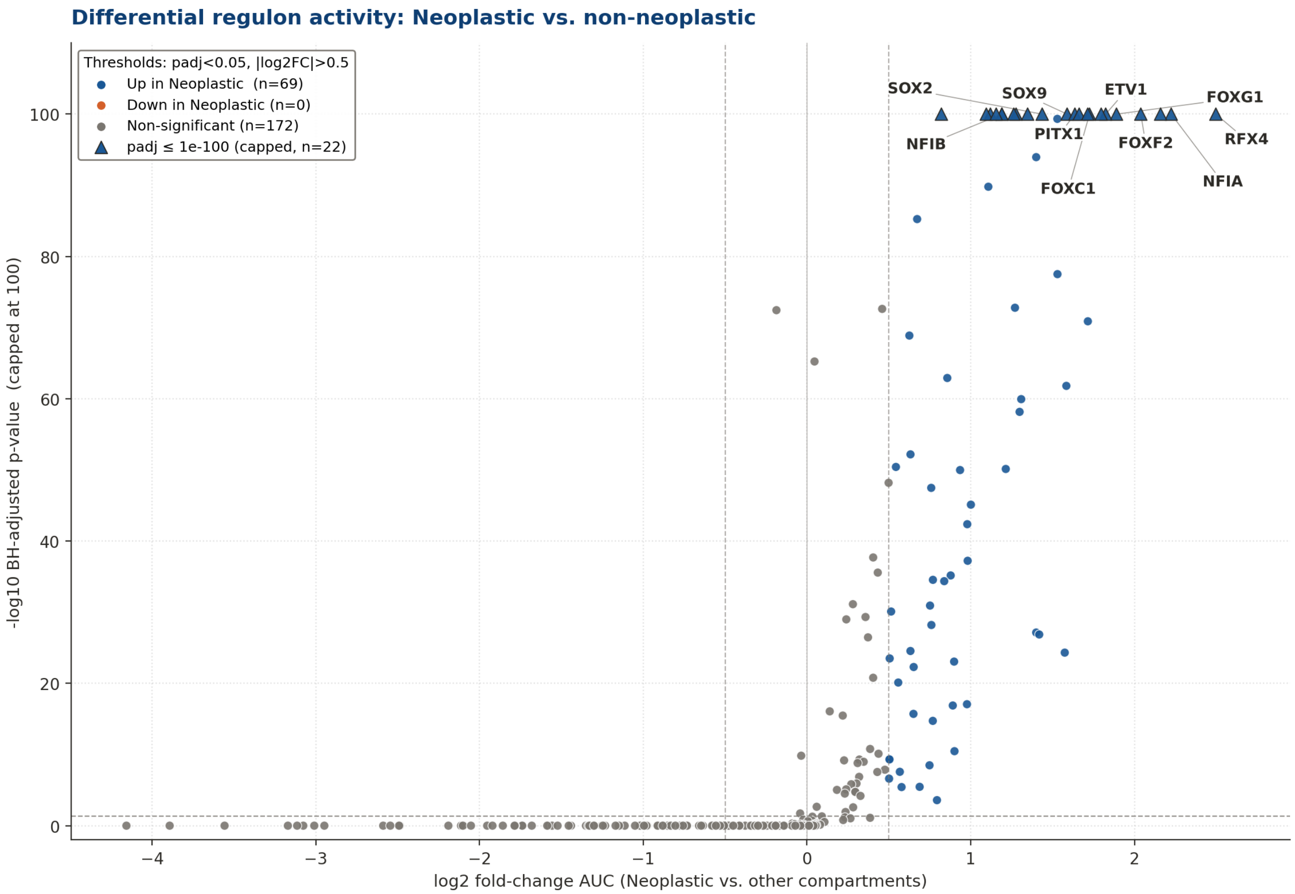

__Figure 8.__ Volcano plot of differential regulon activity (Neoplastic vs. non-neoplastic). X-axis: log2 fold-change of mean AUCell activity; y-axis: -log10 BH-adjusted Wilcoxon p-value (capped at 100 for display). Triangles mark regulons whose raw p-value falls below the cap. Top-10 regulons by combined effect × significance are labelled. Thresholds: padj < 0.05 (horizontal dashed line) and |log2FC| > 0.5 (vertical dashed lines).

3.7 Top regulons per cell type

Table 2 lists the top 5 regulons for each cell type, ranked by Regulon Specificity Score (RSS). These constitute a compact, interpretable catalog of cell-type identity TFs that can be used as starting points for functional follow-up, biomarker design, or therapeutic target prioritization.

Cell type	n cells	Rank	Regulon	RSS
Neoplastic	1034	1	RFX4(+)	0.592
Neoplastic	1034	2	NFIA(+)	0.580
Neoplastic	1034	3	SOX9(+)	0.558
Neoplastic	1034	4	PITX1(+)	0.544
Neoplastic	1034	5	SOX2(+)	0.538
Immune cell	1801	1	IRF8(+)	0.793
Immune cell	1801	2	REL(+)	0.790
Immune cell	1801	3	RUNX3(+)	0.766
Immune cell	1801	4	MAFB(+)	0.761
Immune cell	1801	5	ETS2(+)	0.758
OPC	400	1	SOX21(+)	0.566
OPC	400	2	SOX8(+)	0.551
OPC	400	3	SOX10(+)	0.536
OPC	400	4	ETV5(+)	0.466
OPC	400	5	KLF12(+)	0.456
Astrocyte	82	1	PRDM16(+)	0.370
Astrocyte	82	2	ZNF214(+)	0.328
Astrocyte	82	3	OTX1(+)	0.316
Astrocyte	82	4	LHX2(+)	0.299
Astrocyte	82	5	PAX6(+)	0.283
Oligodendrocyte	75	1	HOXD1(+)	0.331
Oligodendrocyte	75	2	ZNF721(+)	0.294
Oligodendrocyte	75	3	NR2C2(+)	0.281
Oligodendrocyte	75	4	MAFG(+)	0.261
Oligodendrocyte	75	5	RFXANK(+)	0.258
Vascular	47	1	FOXC1(+)	0.323
Vascular	47	2	FOXF2(+)	0.302
Vascular	47	3	GATA6(+)	0.290
Vascular	47	4	PRDM1(+)	0.283
Vascular	47	5	ETS1(+)	0.263
Neuron	20	1	DLX2(+)	0.385
Neuron	20	2	DLX6(+)	0.353
Neuron	20	3	DLX5(+)	0.281
Neuron	20	4	DLX1(+)	0.257
Neuron	20	5	ZNF354A(+)	0.214

Table 2. *Top 5 regulons per cell type, ranked by Regulon Specificity Score (RSS, Jensen–Shannon-based). Cell types with $n < 30$ (Neuron) are reported for completeness but were excluded from the differential test.*

4. Discussion

The pySCENIC analysis recovers a biologically coherent and literature-consistent regulatory map of glioblastoma. Three observations are particularly noteworthy.

4.1 Recovery of canonical GBM master regulators

The neoplastic compartment is anchored by **SOX2 and SOX9** — both are established master regulators of glioma stem-cell identity (Suvà 2014; Glasgow 2014). The presence of NFIA and RFX4 as top neoplastic regulators is consistent with the developmental-hijack model: NFIA drives gliogenic specification in neural stem cells [7], and RFX4 marks ventricular-zone neural progenitors during cortical development. Their re-activation in tumor cells supports the view that GBM cells co-opt embryonic neural-progenitor programs. PITX1 is less canonical and may be a candidate for follow-up validation.

4.2 Tumor microenvironment is dominated by myeloid cells

The immune compartment (~52% of all cells) is overwhelmingly myeloid: the top regulons **IRF8**, **RUNX3**, **MAFB**, and **REL** are textbook microglia/macrophage TFs. This re-confirms previous findings that GBM TME is dominated by tumor-associated macrophages and microglia rather than lymphocytes [8] — the typical immunologically 'cold' phenotype that motivates ongoing myeloid-targeting therapeutic strategies (e.g., CSF1R inhibitors).

4.3 OPC-like cells form a distinct regulatory state

OPCs occupy a clearly distinct regulatory state driven by **SOX10, SOX8, and SOX21**. Several studies have proposed an OPC-like malignant cell state that contributes to GBM heterogeneity (Neftel 2019 [9]); our cell-type-specific recovery of SoxE family regulons in OPCs (vs. SOX2/SOX9 in neoplastic cells) provides clean regulon-level discrimination between non-malignant OPCs and OPC-like tumor cells.

4.4 Caveats

- **Sample size at the patient level (n=4)** is too small to support robust patient-level inferences; all results are reported at the cell-type rather than patient level.
- **Smart-seq2 vs. 10x**: this dataset is Smart-seq2 (deeper per-cell sequencing, lower throughput). pySCENIC was originally developed and is most extensively validated on 10x Genomics data. The cisTarget databases (hg38, v10) and GRN inference work equally on Smart-seq2, but per-cell AUCell scores tend to be sharper here than in 10x.
- **Single GRNBoost2 run** (Standard rigor): we did not run the multi-seed stability subsampling protocol from the SCENIC stability extension. As such, individual TF→target edges should be considered hypotheses; the cell-type-specific *regulon*-level conclusions (which aggregate across many targets) are robust.
- **Author annotations**: cell-type labels are inherited from Darmanis et al.'s original annotation; the 'Neoplastic' label combines all malignant cells without subclassifying into mesenchymal/proneural/classical states.

- **Motif evidence, not ChIP-seq**: cisTarget regulons are supported by motif enrichment, not by direct ChIP-seq binding. Targets are putative; experimental validation is required for individual TF→target claims.
- **Reference genome mismatch**: counts are hg19-aligned (original publication) but cisTarget v10 uses hg38. Because both work on HGNC symbols rather than coordinates, this is compatible; ~99% of expressed gene symbols are stable across hg19 and hg38.

5. Conclusions

Single-cell pySCENIC analysis of 3,459 GBM cells from 4 patients recovers 241 motif-supported regulons that cleanly discriminate the major cellular compartments of glioblastoma and its microenvironment. Each compartment is anchored by literature-canonical transcription factors (SOX2/SOX9 in neoplastic cells, SOX10 in OPCs, IRF8/MAFB in microglia, FOXC1/GATA6 in vasculature). The resulting cell × regulon matrix is a substantially less noisy, biologically interpretable alternative to raw gene expression for downstream analyses such as patient stratification, target prioritization, and drug-response modelling. The pipeline is reproducible at standard rigor and ready for extension to larger GBM cohorts or patient-stratified comparisons.

6. References

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